Interactive TDRP diagnostics and evolutionary calibration



A quick guide to DeepPoW

Presented By DeepPoW Development Team

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I Launch DeepPoW — Installation & Startup

Step 1: Prepare R Environment

- Install R (≥ 4.2.0) and the required R packages.
- Use the dependency installation instructions provided in the repository.

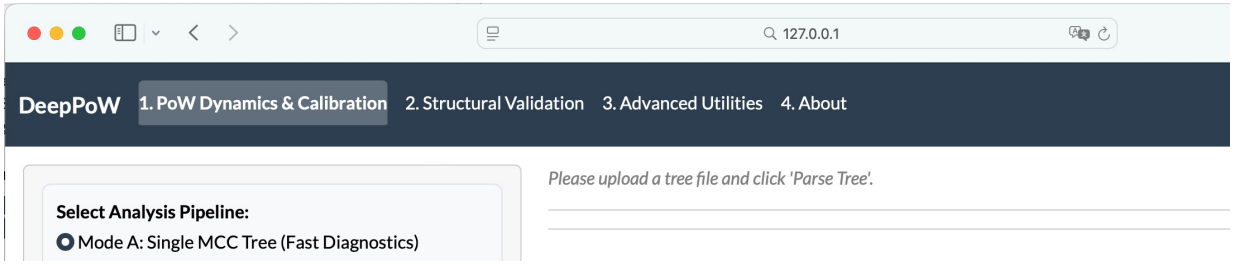
Step 2: Launch the Application

- Open the DeepPoW project in R/RStudio and run the main Shiny application.
- Run from the application directory; confirm the exact entry point in the repository.



Step 3: Verify Successful Startup

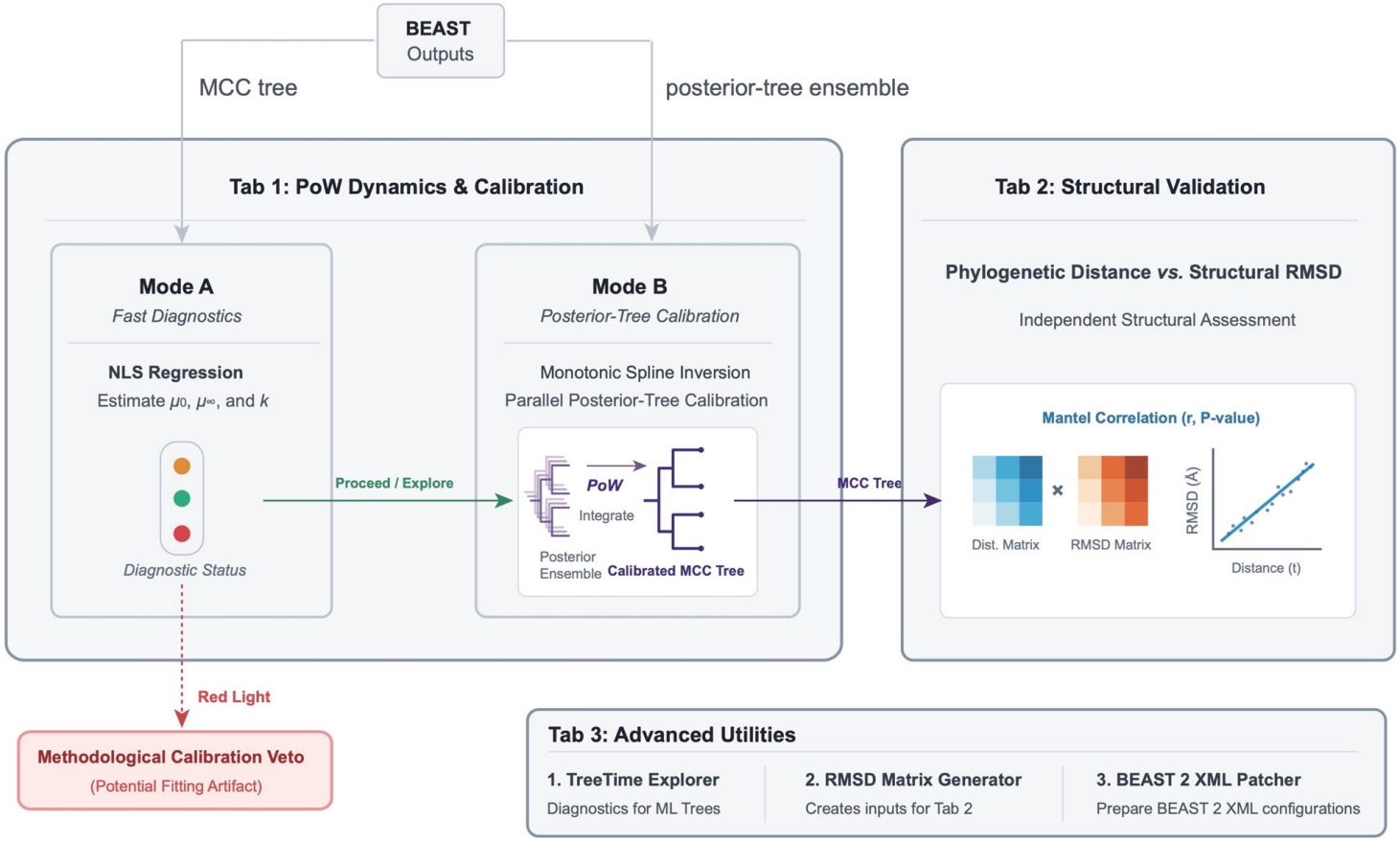
- Confirm that the DeepPoW interface opens and its main tabs are accessible.





For detailed installation instructions and dependency requirements, visit the GitHub repository: https://github.com/BioEasy/DeepPoW.

II DeepPoW Workflow



Important **Mode A** and **Mode B** use separate tree inputs. A Red Light is a methodological calibration veto, not a software-enforced lockout.

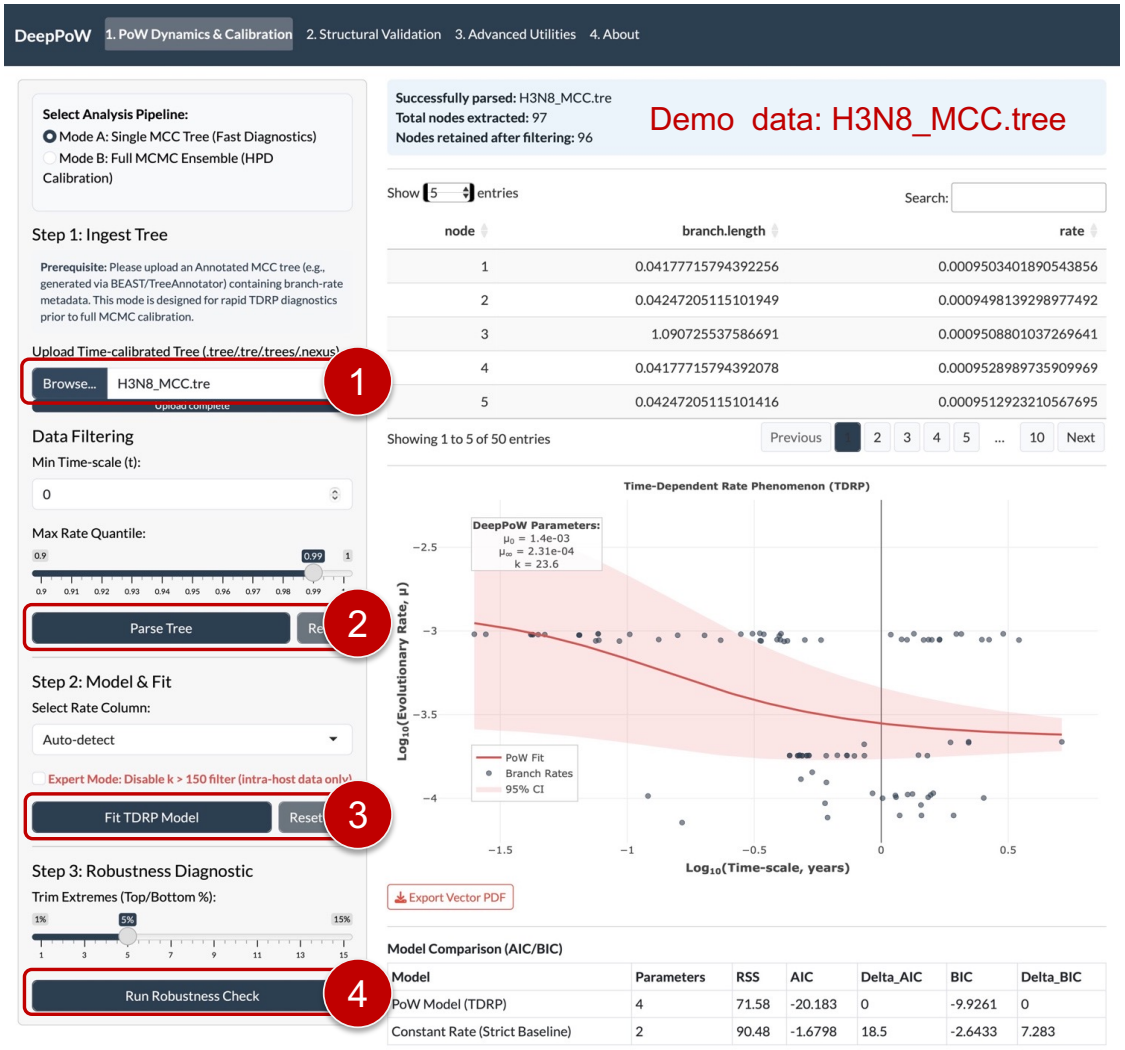
III Core modules & Inputs / outputs

Module	Required / typical input	Main output
Mode A	BEAST time-scaled MCC / summary tree	μ_0, μ_∞, k; fit, AIC/BIC, Traffic Light
Mode B	BEAST posterior .trees + model settings	PoW-calibrated posterior .trees
Structural Validation	Calibrated annotated tree + RMSD CSV	Mantel *r*, permutation *P*-value
TreeTime Explorer	Time-scaled ML tree (.nwk / .nexus)	Exploratory TDRP fit and diagnostic
RMSD Matrix Generator	Matched protein structures (.pdb / .cif)	Pairwise RMSD matrix (CSV)
BEAST 2 XML Patcher	Supported BEAST 2 XML	Patched XML; annotated MCC tree (after BEAST 2 and TreeAnnotator)

Note: μ_0, initial rate parameter; μ_∞, long-term substitution rate; and k, decay coefficient.

IV Demo | Mode A (Fast Diagnostics)

Tab 1: PoW Dynamics & Calibration



Step-by-step operation

1. **Upload Time-calibrated Tree** — Load a time-scaled BEAST summary tree.
2. **Parse Tree** — Parse tree and branch-rate information.
3. **Fit TDRP Model** — Estimate μ_0, μ_∞, and k.
4. **Run Robustness Check** — Assess parameter stability.

Results

- Review the fitted curve, model comparison, and **Traffic Light** status.

Robustness Diagnostic Results

Parameter	Full_Data	Trimmed_5pct	Shift_Percent
μ_0 (Initial Rate Parameter)	0.0014	0.001386	1.00%
$\mu\infty$ (Long-term Substitution Rate)	0.0002308	0.000232	0.51%
k (Decay Coefficient)	23.61	21.98	6.89%

Globally Robust: The decay coefficient (k) shifted by 6.9% after removing outliers. The TDRP signal is highly stable and shows limited sensitivity to the trimming procedure.

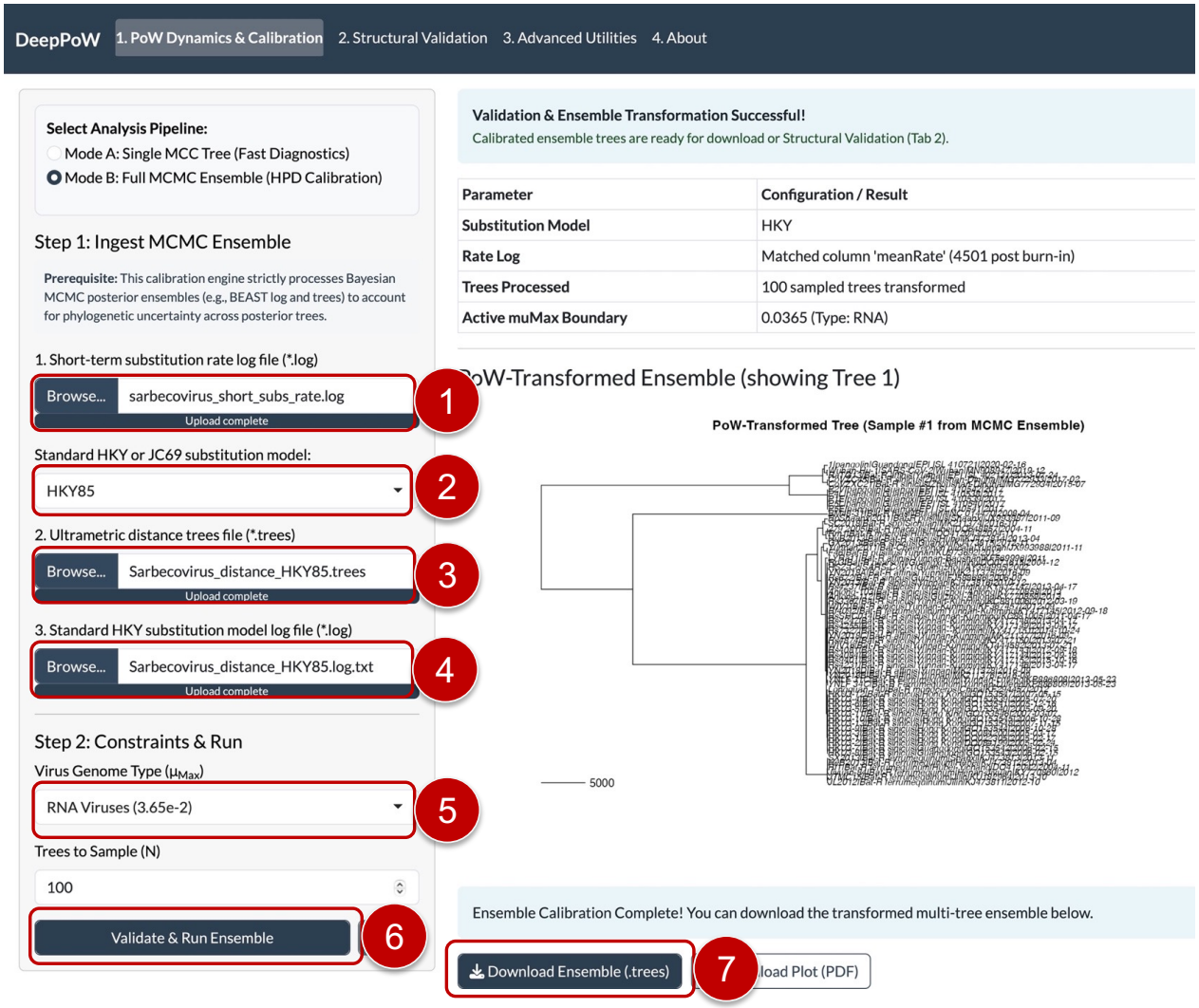
Tree Calibration (Robustness Verified)

A robust TDRP pattern consistent with time-dependent rate decay was detected.

→ Please proceed to Mode B to perform rigorous MCMC ensemble calibration. Single tree extraction is disabled for MCC trees to preserve HPD intervals.

Demo | Mode B (Posterior-Tree Calibration)

Tab 1: PoW Dynamics & Calibration

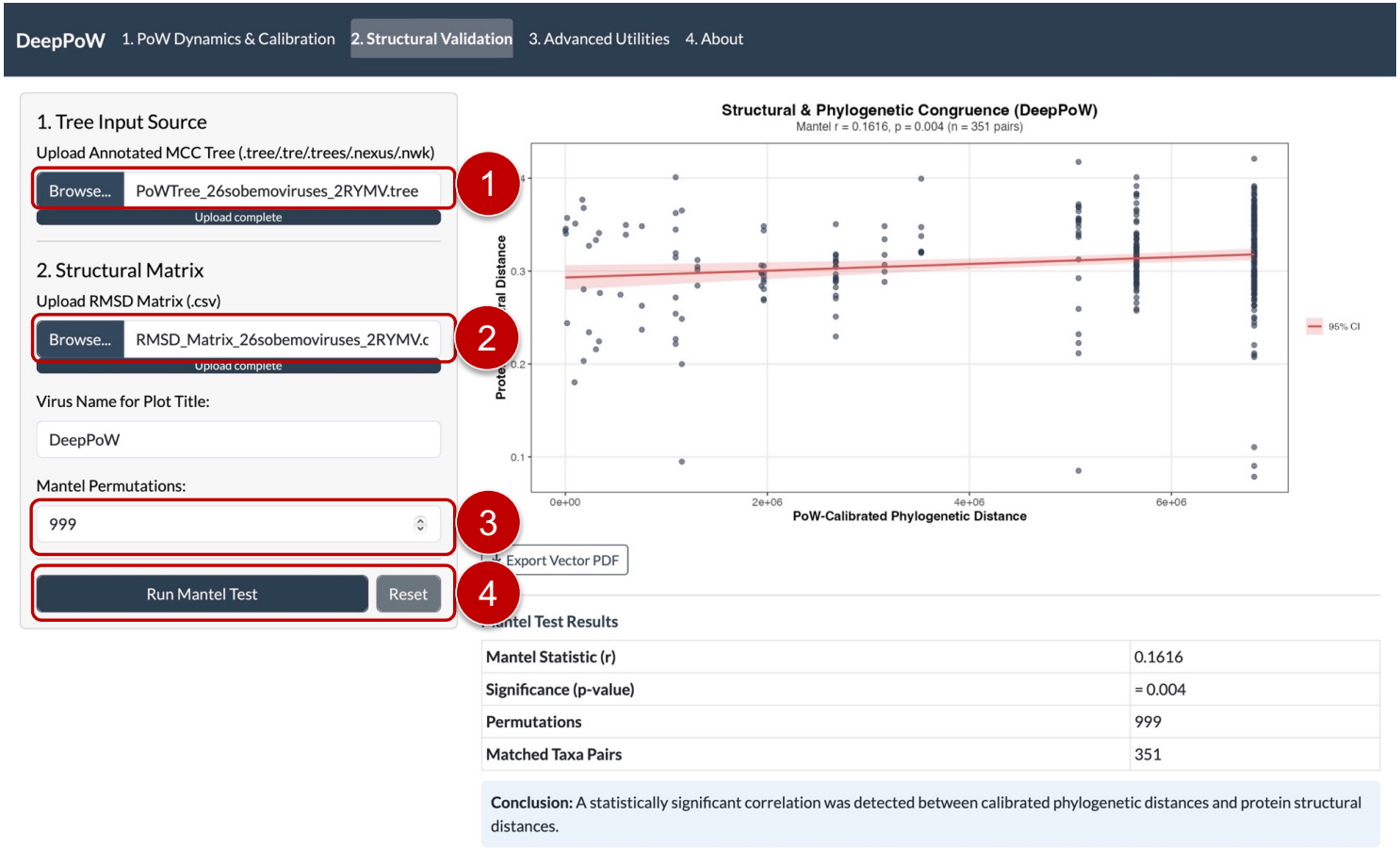


Step-by-step operation

1. **Upload Rate Log** — Load the short-term substitution-rate log file (.log).
2. **Select Substitution Model** — Choose HKY85 or JC69 substitution model.
3. **Upload Distance Trees File** — Load the ultrametric distance trees file (.trees).
4. **Upload HKY Log** — Provide substitution-model parameters (HKY85 only).
5. **Set Constraints** — Configure μ_{max} and tree sampling.
6. **Validate & Run Ensemble** — Execute posterior-tree calibration.
7. **Download Results** — Export the calibrated posterior-tree ensemble.

VI Demo | Structural Validation

Tab 2. Structural Validation



Step-by-step operation

1. **Upload Phylogeny** — Load the annotated post-calibration tree.
2. **Upload RMSD Matrix** — Load the structural distance matrix.
3. **Set Permutations** — Specify the number of permutations.
4. **Run Mantel Test** — Assess phylogenetic–structural correspondence.

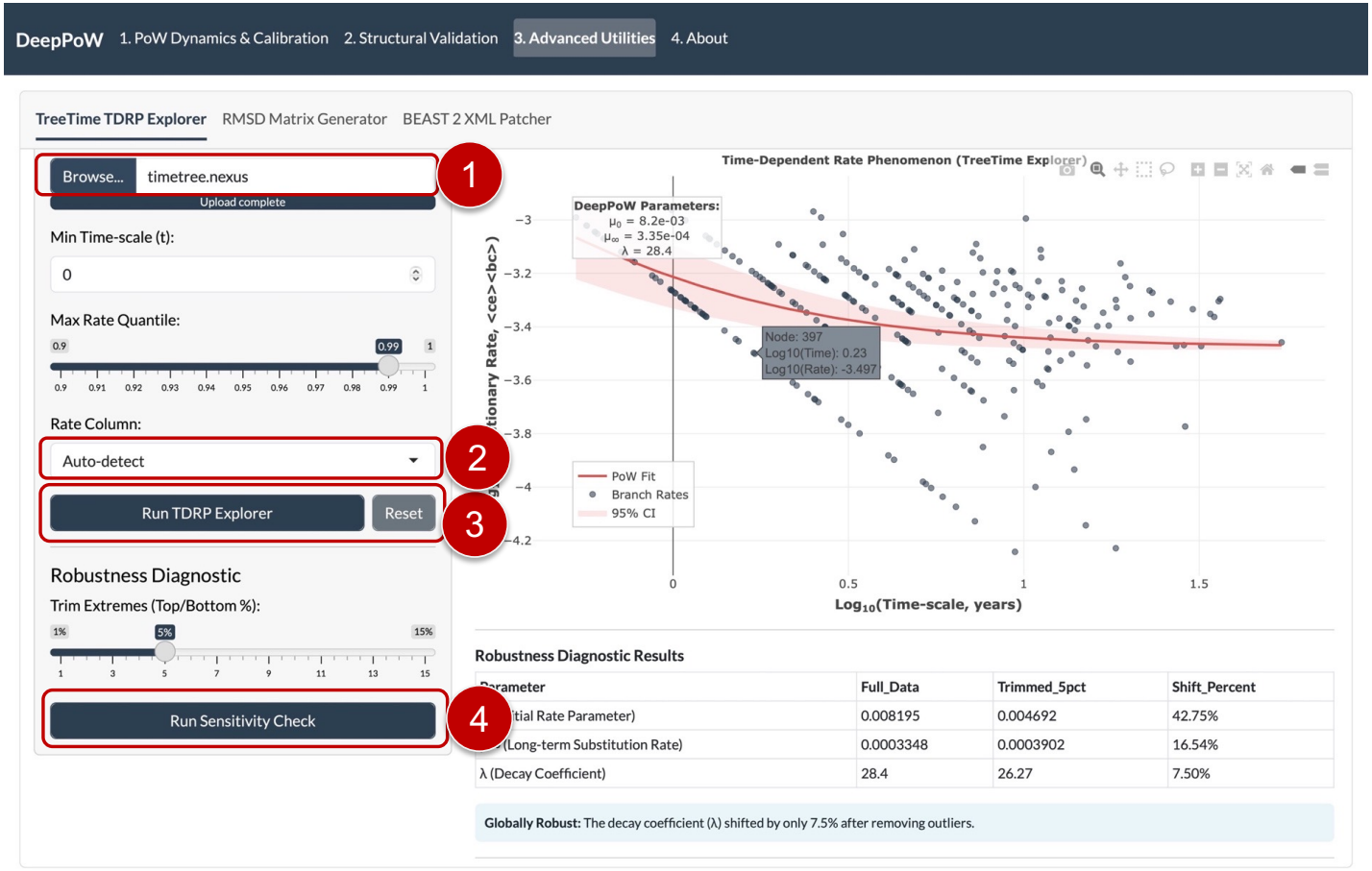
Results

- Review Mantel *r*, permutation *P*-value, and test settings.

Ensure that **taxon names match** exactly between the phylogenetic tree and the RMSD matrix.

VII Demo | TreeTime TDRP Explorer

Tab 3. Advanced Utilities



Step-by-step operation

1. **Upload Tree** — Load a TreeTime-generated .nwk or .nexus tree.
2. **Check Rate Column** — Verify auto-detection or select manually.
3. **Run TDRP Explorer** — Fit the rate-decay model.
4. **Run Sensitivity Check** — Assess parameter stability.

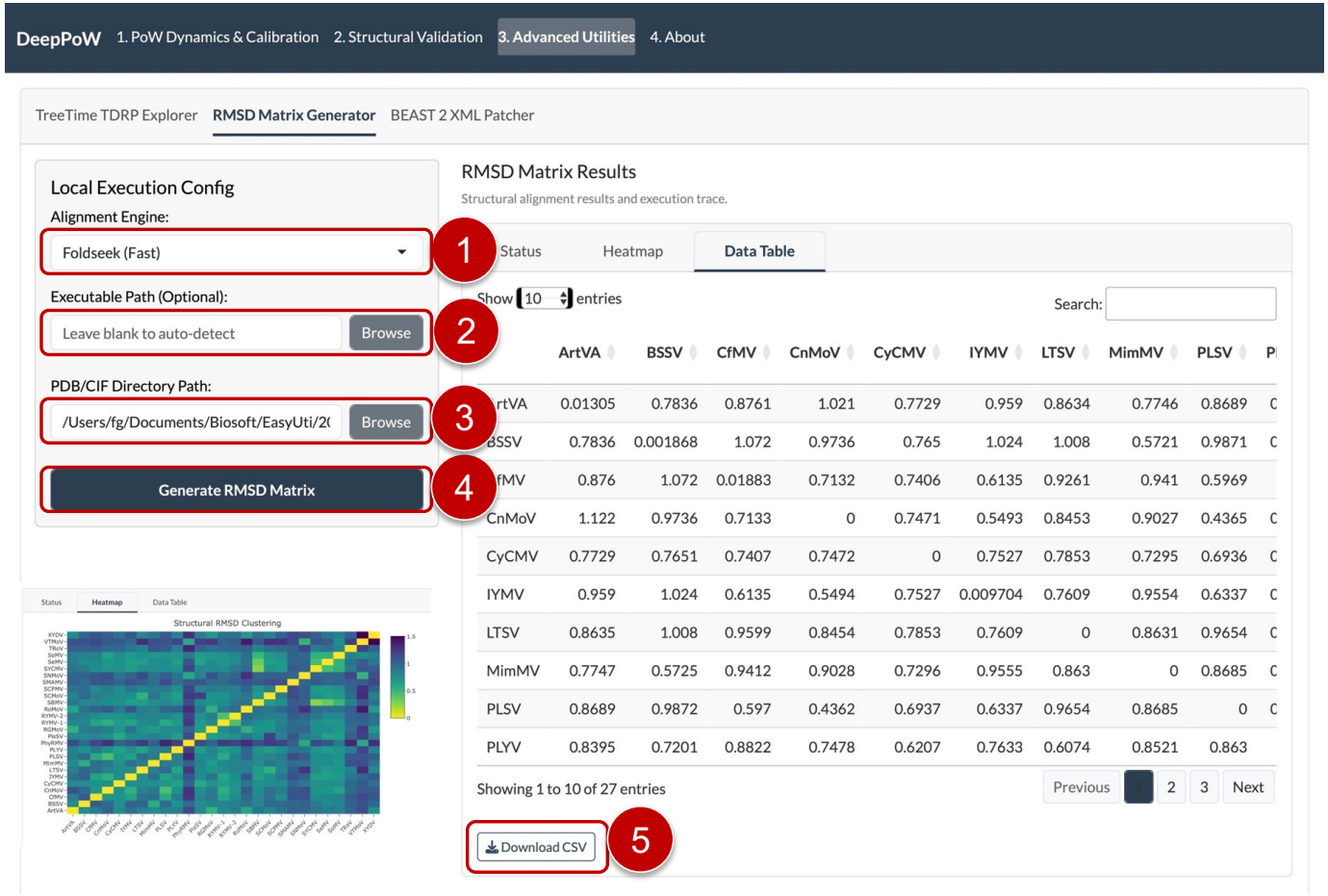
Results

- Review the fitted trajectory and Traffic Light status.

TreeTime Explorer is a preliminary screening tool, **not a substitute for Bayesian posterior calibration**.

VIII Demo | RMSD Matrix Generator

Tab 3. Advanced Utilities



Step-by-step operation

1. **Select Engine** — Choose Foldseek or TM-align.
2. **Check Engine PATH** — Leave blank to auto-detect; if unsuccessful, select the executable from the **plugin** directory.
3. **Upload Structures** — Load multiple .pdb or .cif files.
4. **Generate RMSD Matrix** — Calculate pairwise structural distances.
5. **Export Results** — Save the RMSD matrix as CSV.

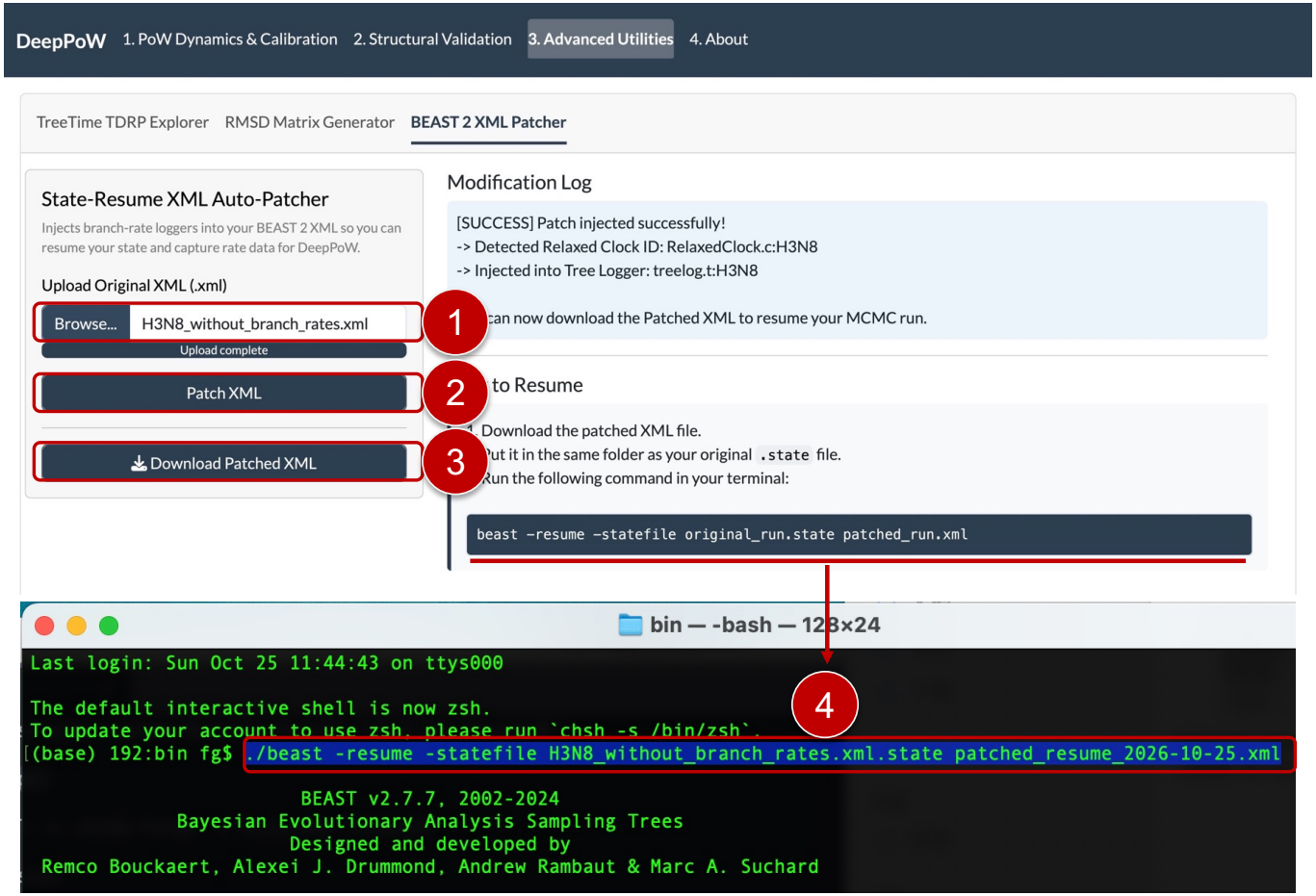
Results

- Review the execution trace and export the RMSD matrix (.CSV) or structural heatmap (.PDF).

Ensure structure filenames match the corresponding tree taxon identifiers exactly.

IX Demo | BEAST 2 XML Patcher

Tab 3. Advanced Utilities



Step-by-step operation

1.Upload XML — Load the BEAST 2 XML file to be patched.

2.Patch XML — Apply the supported XML modifications.

3.Download Patched XML — Export the modified BEAST 2 XML file.

4.Run Patched XML with Statefile — Run BEAST 2 using the patched XML and the corresponding statefile.

Results

- Generate an MCC tree with branch-specific substitution-rate annotations for subsequent DeepPoW Mode A diagnostics.



After the BEAST 2 run, use **TreeAnnotator** to summarize the posterior trees into an annotated MCC tree.

X Troubleshooting

Symptom	First check	Next action
Tree fails to load	Nexus/Newick format and annotations	Re-export tree with required branch-rate data
Unexpected large *k*	Short terminal branches and sampling density	Inspect outliers; review inter-host *k* > 150 warning
Red Light / unstable fit	*k* shift after 5% + 5% trimming	Investigate influential branches before routine calibration
Mode B fails	Posterior file, model settings, ESS	Verify MCMC convergence and annotations
Foldseek / TM-align missing	Executable and PATH in R environment	Activate environment; restart R/RStudio
RMSD / Mantel problem	Taxon labels, matrix quality	Check names, diagonal and structure correspondence
Patched XML fails	BEAST 2/package compatibility	Inspect XML; perform short test run



see **User Manual** for detailed causes and remedies.

XI Quick reference

1. Prepare	Quality-check sequences, dates, annotations and MCMC.
2. Diagnose	**Mode A**: estimate μ_0, μ_∞, k and robustness shift.
3. Decide	Interpret Green / Yellow / Red; investigate cautions.
4. Calibrate	**Mode B**: transform posterior trees if scientifically justified.
5. Validate	Optional RMSD/Mantel comparison and transparent reporting.



DeepPoW is an analysis aid, not a substitute for careful evolutionary interpretation.

For reproducibility: Record the DeepPoW version, input files, key analysis settings, and exported results.

XII Contact us

DeepPoW is an ongoing project

We welcome bug reports, feedback, and suggestions. **E-mail**: raindy@fafu.edu.cn or swayzhang@fafu.edu.cn

Project & documentation

https://github.com/BioEasy/DeepPoW

Read the **User Manual** for methodological detail and supported input formats.

Bug reports

Open an issue in the GitHub repository. Include version, OS, R session information, steps to reproduce, and a minimal non-sensitive example.

Before submitting

Check existing issues; remove private sample identifiers; attach console warnings and relevant screenshots. For scientific questions, cite the **manuscript** and **Manual**.